

Dean Kuhn

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SUMMARY

Computer science graduate (WGU, May 2026) focused on data engineering and pipeline development. Built and deployed end-to-end data projects on a live Postgres/Snowflake/dbt warehouse stack: an ML demand-forecasting system with automated A/B evaluation and live Power BI and Streamlit reporting, and a weekly ingestion pipeline powering longitudinal growth analysis for 7,700+ artists. Seeking a junior data engineering, analytics engineering, or data analysis role.

PROJECTS

[KitchenSync Food Forecasting System](#) · 2026

Tech: Python · FastAPI · PostgreSQL (Neon) · Snowflake · dbt Core · LightGBM · Streamlit · Power BI · AWS EC2

- Built an end-to-end demand forecasting system for a simulated 12-store retail kitchen chain, replacing static production guesswork with **per-store, per-item forecasts at 15-minute grain**: live POS event ingestion (FastAPI → Postgres), nightly extract to Snowflake, three-layer dbt transformation, LightGBM model, and a Streamlit dashboard serving live production queues to kitchen staff.
- Proved model value with a nightly A/B comparison against an hourly-average baseline under identical seeded demand: **338,688 weekly predictions cut stockouts ~40% and lifted service level +1.6pp**, at a measured cost of +3.3pp additional waste — a real trade-off surfaced, not a cherry-picked result.
- Diagnosed **two silent data-quality bugs** via root-cause debugging: a day-of-week convention mismatch between Snowflake EXTRACT and ISO weekday numbering zeroed **2.37M training rows** (producing a near-constant model), and a conditional-mean bias that inflated demand predictions 3–4x at low-traffic stores.
- Built a **3-page Power BI dashboard connected live to Snowflake**: a store waste leaderboard using RANKX so region slicers can't silently re-rank only the visible subset, a drillthrough-only store detail page, and synced category-filtered top-seller and worst-waste-cost charts.
- Deployed the full system on AWS EC2 (systemd services with auto-restart); a nightly cron chains extract → dbt → predict → A/B comparison → publish, and GitHub Actions rebuilds the portfolio site with fresh results each morning.

[Music Growth Pipeline](#) · 2026

Tech: Python · PostgreSQL (Neon) · dbt Core · Last.fm API · GitHub Actions · SQL · Power BI

- Built the longitudinal dataset the Last.fm API doesn't provide: the API returns only cumulative all-time stats, so a weekly automated pipeline snapshots listener and playcount data for **7,751 artists** into a cloud Postgres warehouse, enabling week-over-week growth analysis across mainstream and indie tiers.
- Surfaced the core finding across 7 weeks of data: underground artists (chart pages 1000+) show **P90 listener growth of 9.16% vs. 2.75% for mainstream** — a **3x gap** concentrated in the tail; fresh stats publish automatically to the portfolio site after each weekly run.
- Built a **two-page Power BI report connected live to the Postgres warehouse**: growth trends by tier and genre, plus an artist-level drillthrough showing individual listener timelines — aggregate and per-artist views with no batch export step.
- Modeled the transformation layer in dbt (6 staging + 7 mart models; marts build on marts via ref(); LAG window functions) and extended the schema with genre associations (15 genres × 500 artists) and artist similarity networks (~2,000 artists).

[VRP-TW Genetic Algorithm Routing Optimizer](#) · 2026

Tech: Python · Genetic Algorithm · CLI

- Built a Genetic Algorithm that optimizes package-to-truck assignment and delivery sequencing **simultaneously** via a sentinel chromosome encoding — one fitness function instead of two sequential passes over an NP-hard routing problem.
- Engineered convergence controls: adaptive mutation (rate doubles after 50 stagnant generations) and a gradient lateness penalty (minutes_late × 10) creating a smooth fitness landscape over binary pass/fail.

TECHNICAL SKILLS

Languages: Python, SQL

Data & Storage: PostgreSQL, Snowflake, dbt Core (dbt-postgres, dbt-snowflake), pandas, PyArrow, Parquet

Visualization: Power BI, Streamlit, Plotly

Cloud & Infra: AWS EC2, GitHub Actions, Docker, Neon (cloud Postgres), systemd, Git, Linux/WSL

Ingestion & APIs: FastAPI, REST APIs, asyncio/httpx, Playwright (headless scraping)

ML & Modeling: LightGBM, scikit-learn

Concepts: ETL/ELT pipeline design, medallion architecture, data warehouse modeling, dbt staging/mart patterns, ML inference serving

Exposure: PySpark, Apache Airflow, Databricks, Azure Data Lake Storage, Delta Lake

EDUCATION

B.S. Computer Science · Western Governors University · May 2026

WORK EXPERIENCE

Guest Services & Inventory Coordinator · Kwik Trip · 2021 – Present

- Tracked inventory and managed ingredient purchasing for kitchen operations in a high-volume food production environment (the same system KitchenSync models), balancing stock daily to avoid both shortages and excess waste.
- Ran end-of-shift cash reconciliation and register audits across high daily transaction volume; caught and resolved discrepancies before close.
- First-line troubleshooting for POS terminal failures, network issues, and delivery problems during live shifts with no dedicated IT support on site.